Challenge for "Merit Prize 2025-26"

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1 Introduction

This project addresses the task of detecting fake news in Portuguese using machine learning models trained on the provided dataset. We evaluate several classifiers under different preprocessing strategies, identify the best-performing model, and use interpretability techniques to understand the linguistic patterns driving predictions. Our analysis highlights the impact of duplicates, lemmatization, and numerical features on performance, and examines both local and global explanations to assess model reliability.

2 Model Training and Evaluation

2.1 Feature Extraction

We employed TF-IDF vectorization to transform text into numerical representations, capturing word importance while accounting for corpus-wide prevalence. Our pipeline evaluated three data partitions (training, validation, test) across multiple preprocessing strategies: original text versus lemmatized forms (reducing inflected words to base forms), and with/without duplicates.

TfidfVectorizer parameters included minimum document frequency of 10, maximum of 90%, vocabulary limit of 5,000 features, Portuguese stopword removal, and lowercase normalization. An ablation study revealed that retaining numeric tokens—initially assumed semantically irrelevant—notably improved accuracy across classifiers. Numbers appear to carry discriminative value as stylistic markers (dates, statistics, quantities) distinguishing authentic journalism from fabricated content in Portuguese news.

2.2 Model Training

We performed hyperparameter optimization using F1-score to balance precision and recall, suitable for binary classification where both error types matter. Tuning was conducted separately for original and lemmatized text configurations.

To ensure robust evaluation and mitigate overfitting, we employed stratified 5-fold cross-validation with shuffling (seed=42 for reproducibility). Stratification maintained consistent class distributions across folds, preserving fake-to-authentic ratios in all splits. Final models were assessed using precision, recall, accuracy, and F1-score to capture comprehensive performance across both classes.

2.2.1 Decision Tree

For Decision Tree optimization, we performed exhaustive hyperparameter tuning across four key parameters that control model complexity and class balance. The explored parameter space included: (i) `max_depth` $\in \{10, 30, 50\}$, (ii) `min_samples_split` $\in \{2, 5, 10, 20\}$, (iii) `min_samples_leaf` $\in \{1, 2, 4\}$, and (iv) `class_weight` $\in \{None, Balanced\}$.

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Test Dataset	max_depth	min_samples_split	min_samples_leaf	class_weight
Not lemmatized, with duplicates	Balanced	50	1	2
Lemmatized, with duplicates	Balanced	50	1	5
Lemmatized, without duplicates	None	30	2	10

Table 1: Best Decision Tree Parameters

2.2.2 Gaussian Naive Bayes

For Gaussian Naive Bayes optimization, we tuned the variance smoothing parameter (`var_smoothing`) across the range $\{10^{-9}, 10^{-8}, 10^{-7}, 10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1.0\}$.

Test Dataset	var_smoothing
Not Lemmatized, with duplicates	1e-09
Lemmatized, with duplicates	1e-07
Lemmatized, without duplicates	1e-03

Table 2: Best Gaussian Naive Bayes Parameters

2.2.3 Logistic Regression with L2 regularization

For L2-regularized Logistic Regression, we tuned four key hyperparameters: (i) inverse regularization strength $\in \{0.01, 0.1, 1, 10\}$, (ii) solver algorithm $\in$ {'lbfgs', 'liblinear', 'saga'}, (iii) maximum iterations $\in \{100, 200, 500, 1000\}$, and (iv) class weight $\in$ {None, 'balanced'}.

(Note: C = `inverse_regularization_strength`)

Test Dataset	C	solver	max_iter	class_weight
Not lemmatized, with duplicates	10	Balanced	100	lbfgs
Lemmatized, with duplicates	10	Balanced	100	liblinear
Lemmatized, without duplicates	1	None	100	liblinear

Table 3: Best Logistic Regression L2 Regularization Parameters

2.2.4 Logistic Regression with L1 regularization

For L1-regularized Logistic Regression, we tuned four key hyperparameters: (i) inverse regularization strength `C` $\in \{0.01, 0.1, 1, 10\}$, (ii) solver algorithm $\in$ {'liblinear', 'saga'}, restricted to L1-compatible optimizers, (iii) maximum iterations $\in \{100, 200, 500, 1000\}$, and (iv) class weight $\in$ {None, 'balanced'}.

Test Dataset	C	solver	max_iter	class_weight
Not lemmatized, with duplicates	1	None	100	liblinear
Lemmatized, with duplicates	10	Balanced	100	saga
Lemmatized, without duplicates	1	None	100	liblinear

Table 4: Best Logistic Regression L1 Regularization Parameters

2.2.5 Multi-Layer Perceptron (MLP)

For Multi-Layer Perceptron (MLP) optimization, we tuned five key hyperparameters: (i) hidden layer architecture $\in \{(50,), (100,), (200,), (300,), (64, 32), (100, 50), (200, 100), (128, 64)\}$, exploring

both single and two-layer configurations, (ii) activation function $\in$ {'logistic', 'tanh', 'relu'}, (iii) optimizer $\in$ {'lbfgs', 'sgd', 'adam'}, (iv) L2 penalty strength $\in \{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}\}$, and (v) learning rate schedule $\in$ {'constant', 'adaptive'}. Training was limited to 500 iterations with early stopping enabled to prevent overfitting by terminating when validation performance plateaued.

Test Dataset	solver	learning rate init	learning rate	hidden layers sizes	alpha	activation
Not lemmatized, with duplicates	adam	0.05	adaptive	`(50,)`	1e-5	logistic
Lemmatized, with duplicates	adam	0.0005	constant	`(64,32)`	1e-3	tanh
Lemmatized, without duplicates	adam	0.001	adaptive	`(200,100)`	1e-2	relu

Table 5: Best MLP Parameters

2.2.6 Best Model and Conclusion

Our results reveal that duplicates substantially improved model performance across all classifiers, with lemmatized datasets achieving the highest accuracies. To compare models side by side, Tables 6–8 summarize accuracy, precision, recall, and F1-score for each preprocessing configuration. These tables allow a clearer understanding of how duplicates, lemmatization, and numerical features affect each classifier's performance.

Decision Tree reached 0.9640 accuracy with duplicates versus 0.8058 without—an 11-point improvement. However, duplicates may artificially inflate metrics through memorization rather than genuine pattern recognition, overestimating real-world performance.

Model	Accuracy	Precision	Recall	F1-Score
Decision Tree	0.8531	0.8535	0.8531	0.8531
Gaussian NB	0.8518	0.8520	0.8518	0.8518
Logistic Regression L2	0.9009	0.9009	0.9009	0.9009
Logistic Regression L1	0.8997	0.8997	0.8997	0.8997
MLP Classifier	0.9086	0.9086	0.9086	0.9086

Table 6: Model Performance Comparison: Not lemmatized, with duplicates

Model	Accuracy	Precision	Recall	F1-Score
Decision Tree	0.9640	0.9643	0.9640	0.9640
Gaussian NB	0.8637	0.8638	0.8637	0.8637
Logistic Regression L2	0.9108	0.9108	0.9108	0.9108
Logistic Regression L1	0.9037	0.9037	0.9037	0.9037
MLP Classifier	0.9203	0.9203	0.9203	0.9203

Table 7: Model Performance Comparison: Lemmatized, with duplicates

Model	Accuracy	Precision	Recall	F1-Score
Decision Tree	0.8058	0.8058	0.8058	0.8058
Gaussian NB	0.8395	0.8408	0.8395	0.8397
Logistic Regression L2	0.8820	0.8830	0.8820	0.8816
Logistic Regression L1	0.8834	0.8840	0.8834	0.8831
MLP Classifier	0.8883	0.8885	0.8883	0.8880

Table 8: Model Performance Comparison: Lemmatized, without duplicates

Multi-Layer Perceptron demonstrated robust performance, achieving 0.9203 accuracy on lemmatized data with duplicates and maintaining the highest duplicate-free accuracy at 0.8883, suggesting

superior generalization. Lemmatization showed variable impact by model, while numeric feature retention consistently improved classification. Numbers appear to carry discriminative information (dates, statistics, quantitative claims) distinguishing fake from authentic Portuguese news.

MLP emerges as the most effective classifier, though duplicate-free metrics provide more realistic generalization estimates for production deployment.

To further evaluate the best-performing MLP model, Figure 1 (page 4) shows its ROC curve, which plots True Positive Rate against False Positive Rate across different classification thresholds. The curve lies well above the diagonal baseline, indicating strong discriminative ability. The corresponding Area Under the Curve (AUC) confirms that the MLP reliably ranks real news above fake news regardless of the chosen threshold, reinforcing the robustness observed in the previous tables.

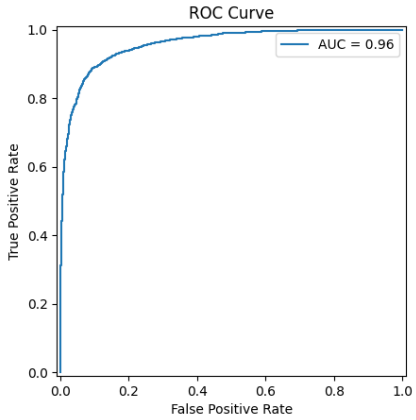


Figure 1: The ROC Curve for the MLP model

3 Model Interpretation

This section analyzes the decision patterns of the best-performing models using feature coefficients, LIME, and permutation importance. Together, these methods provide both local and global insight into how linguistic patterns distinguish fake from real news.

3.1 Top Predictive Words for News Authenticity

The feature coefficients from the best L1-regularized Logistic Regression model provide insight into the terms most associated with each class. Figure 2 displays the top 10 words with the highest positive weights (Real News) and the top 10 with the most negative weights (Fake News).

Notably, fake-news-associated terms display coefficients with magnitudes approaching –20, while the strongest coefficients for real news reach roughly +15. This asymmetry indicates that certain fake-news markers exert a stronger influence on the model's predictions than the features associated with real news.

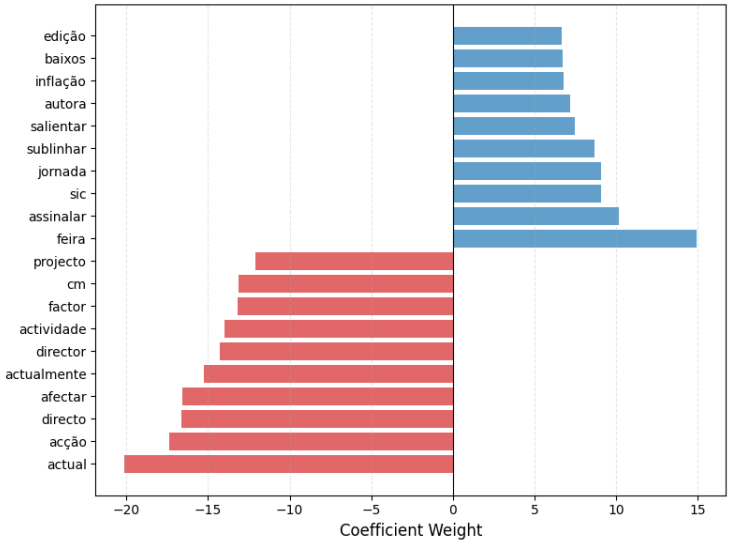


Figure 2: Top 10 Features for Fake (red) vs Real (blue) News Classification

3.1.1 Feature Selection Properties: L1 vs L2 Regularization

L1 and L2 regularization control complexity differently. L1 penalizes absolute coefficient magnitudes, producing sparsity by driving many weights to zero; L2 shrinks coefficients smoothly without eliminating features. In our results, the L1 model retained 2,116 non-zero coefficients (out of 5,000), while the L2 model kept all features.

3.1.2 Selection Criteria: When to Prefer L1 vs L2

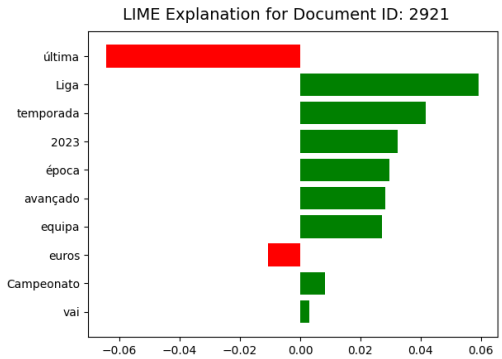
The choice between L1 and L2 regularization depends on the modelling objective.

L1 is preferable when interpretability, sparsity, and dimensionality reduction are priorities, especially in high-dimensional TF-IDF spaces. L2 is more suitable when maximizing predictive performance, handling multicollinearity, or ensuring smooth weight distributions. Although both performed similarly here, L1 provided superior transparency.

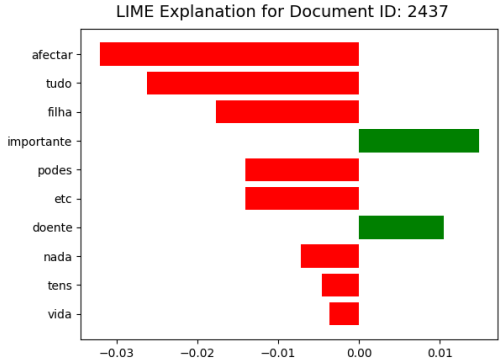
3.2 LIME-Based Interpretation of Logistic Regression Predictions

Local Interpretable Model-agnostic Explanations (LIME) were employed to provide transparent, instance-level justifications for the predictions made by the best Logistic Regression model. This approach generates sparse, local linear approximations of the complex model, highlighting the features (words) that contribute most significantly to a specific document's classification as either Real or Fake. The analysis was performed on samples 2921, 2437, 5557, and 1697 from the validation set, using the L1-regularized Logistic Regression model that achieved an F1-Score of 0.8831 on the lemmatized, duplicate-free dataset.

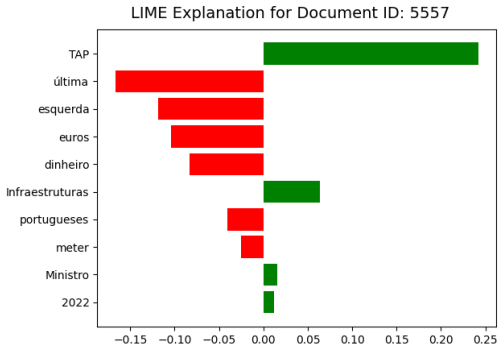
The visualizations generated by LIME clearly depict the relative importance and directional influence of the top features for each prediction. In the resulting bar plots, green bars indicate features supporting the Real News class, while red bars represent features supporting the Fake News class.



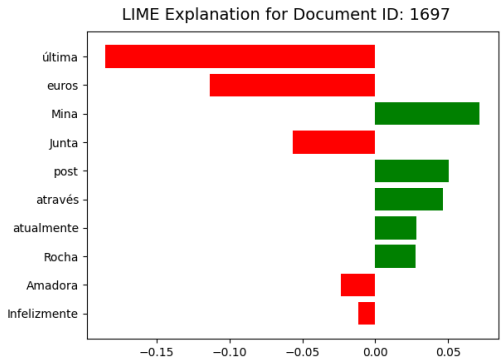
True Class: Real; Probability (Real): 0.9987



True Class: Fake; Probability (Real): 0.0062



True Class: Fake; Probability (Real): 0.0729



True Class: Real; Probability (Real): 0.1229

Figure 3: LIME Explanation for the best logistic regression model predictions of documents 2921, 2437, 5557 and 1697

3.3 MLP Model Interpretation Through LIME and Permutation Importance

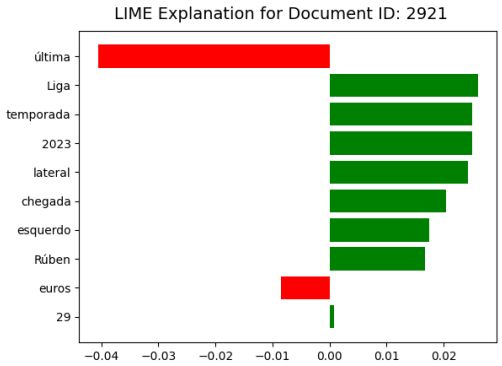
To gain insight into the predictions of the Multi-Layer Perceptron (MLP) model, we utilized two complementary model interpretation techniques: LIME (Local) and Permutation Importance (Global). The MLP model used for this analysis achieved the highest overall accuracy (0.8883) on the duplicate-free, lemmatized test set.

3.3.1 Local Interpretation with LIME

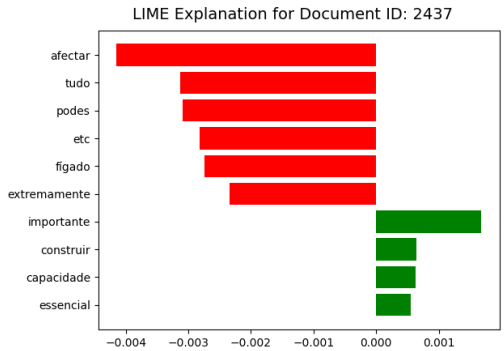
LIME was applied to the same four validation set samples (IDs 2921, 2437, 5557, 1697) to provide instance-level explanations for the MLP's predictions. The MLP, unlike the Logistic Regression model, uses a non-linear combination of features, making LIME crucial for local transparency. The resulting visualizations (Figure 4) illustrate the feature weights for each document's prediction. Green bars indicate features supporting the Real News class, while red bars represent features supporting the Fake News class.

3.3.2 Global Interpretation with Permutation Importance

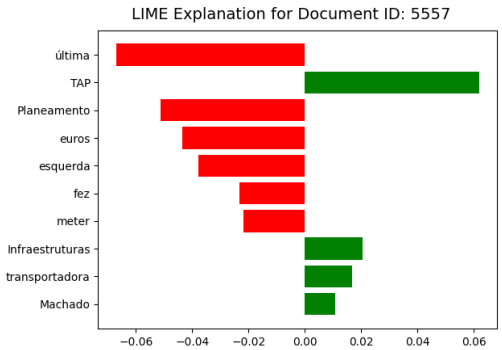
Permutation Importance measures global feature relevance by quantifying the performance degradation that occurs when a feature's relationship with the target is broken. The procedure involves recording a baseline metric (e.g., F1-score) and then randomly shuffling a specific feature while keeping others constant. The resulting drop in performance serves as the importance score; a larger drop indicates the feature is critical for prediction. To account for randomness, we averaged results across 10 permutations ($N_{\text{repeats}} = 10$).



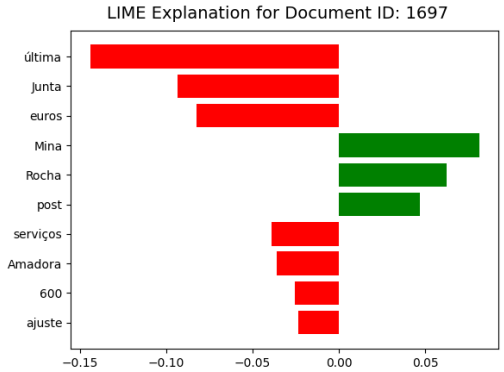
True Class: Real; Probability (Real): 1.0000



True Class: Fake; Probability (Real): 0.0001



True Class: Fake; Probability (Real): 0.0016



True Class: Real; Probability (Real): 0.0206

Figure 4: LIME Explanation for the MLP's predictions of documents 2921, 2437, 5557 and 1697

This analysis was performed using 1,000 random samples from the validation set, measuring the degradation in the F1-score. The results identify the top 20 most important features for the MLP model across the entire subset:

Feature	Permutation Importance	Feature	Permutation Importance
detido	0.005 37	tvi	0.002 67
última	0.004 55	sociais	0.002 58
estados	0.004 33	prr	0.002 58
inflação	0.003 86	causa	0.002 56
segurança	0.003 77	quatro	0.002 51
esquerda	0.003 67	encontro	0.002 39
2021	0.003 24	doenças	0.002 35
feira	0.002 79	fundamental	0.002 31
27	0.002 74	internacional	0.002 31
directo	0.002 73	objectivo	0.002 30

Table 9: Top 20 Features by Permutation Importance for the MLP Model

3.3.3 Discussion: LIME vs. Permutation Importance

This comparison between the two models underscores the trade-off between specificity and generalization. LIME provides local interpretability, highlighting the context-dependent features driving individual predictions, which leads to high variability across documents. In contrast, Permutation Importance offers global interpretability by identifying features that—when neutralized—degrade overall model performance, revealing the most consistently discriminative words across the entire dataset.

4 Clustering

For this part, it was requested to use the K-Means clustering algorithm with `K = 5`, `max_iter = 500`, and `random_state = 42`.

After inspecting the top 3 documents closest to each centroid and analyzing the most representative keywords, we assigned semantic labels to each cluster:

- **Cluster 0: Health & Pandemic Coverage** – This cluster focuses on COVID-19 updates, vaccination policies, and healthcare system discussions. Keywords include *saúde*, *vacina*, *covid*, *médico*, and *hospital*.
- **Cluster 1: Ukraine-Russia War News** – Contains military conflict updates, including contradictory statements from Russian sources and Ukrainian counteroffensive coverage. Representative keywords: *russo*, *ucrânia*, *guerra*, *putin*, *wagner*.
- **Cluster 2: Miscellaneous/General News** – A catch-all cluster with weak cohesion, capturing diverse topics that don't fit well elsewhere. The prevalence of generic verbs (*ter*, *ir*, *fazer*, *poder*) indicates heterogeneous content ranging from economics to sports.
- **Cluster 3: Portuguese Domestic Politics** – Covers Portuguese government affairs, political tensions, and party politics (PS, PSD). Keywords include *ministro*, *governo*, *costa*, *partido*, and *presidente*.
- **Cluster 4: Economic & Fiscal Policy** – Focuses on government spending, fiscal policies, and economic indicators. Representative terms: *euro*, *milhão*, *preço*, *banco*, *taxa*.

It is important to note that the initial hypothesis of these clusters representing "misinformation" categories appears premature. Most documents contain legitimate news coverage on controversial topics rather than verifiable false information. Cluster 2's poor cohesion suggests that increasing the number of clusters or refining preprocessing steps could improve separation.

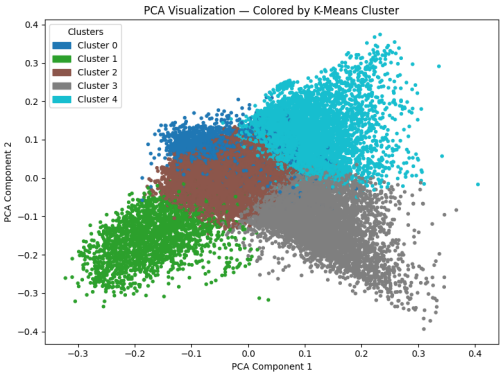


Figure 5: K-Means clustering visualization using PCA dimensionality reduction

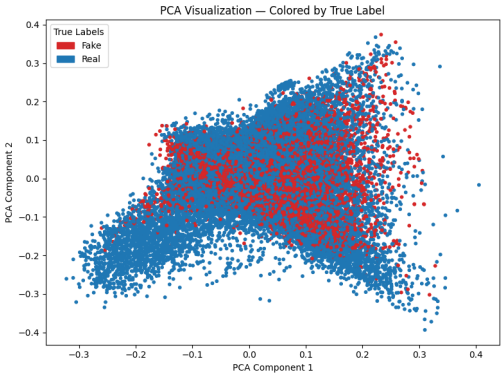


Figure 6: PCA visualization colored by true labels (Fake vs Real news)

Figures 5 and 6 demonstrate that K-Means clustering groups documents by topical similarity rather than veracity. While clear thematic separation emerges (health, politics, war, economics), fake and real articles remain intermixed across clusters, revealing unsupervised clustering's inadequacy for fake news detection.

This occurs because fake news often discusses the same topics as legitimate news but differs in rhetorical strategies, factual accuracy, and linguistic markers. Supervised classification approaches that learn these distinguishing features are therefore more appropriate. Potential improvements include: (1) engineering stylistic features beyond topic keywords (sensationalism, emotional language, credibility markers); (2) semi-supervised learning incorporating known labels; or (3) using clustering as preprocessing for topic-specific models.